Group Theory

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1 Introduction

A lot of mathematics is not just about understanding the objects we use, but the structure behind them and the ways we can manipulate and transform them without changing their essential properties. In algebra, one of the most important structures is called a group. Groups capture the idea of symmetry, which at first sounds very simple; however, after developing the theory of groups, we will see that this simple idea of symmetry is only the starting point of a much deeper algebraic structure that unifies many seemingly unrelated mathematical systems.

2 Groups and examples

Definition 2.1 (Definition of a group)**.** A group is a collection of elements in a set G along with a binary operation $*: G \times G \to G$. For convention, instead of writing $*(a, b)$, we write ab. The set G and the operation $*$ follow the following axioms:

1. Associativity: for any $a, b, c \in G$ $(ab)c = a(bc)$.
2. Identity: There exists an element $e \in G$ such that for all $g \in G$, $ge = eg = g$.
3. Inverses: for any element $g \in G$ there exists another element $h \in G$ such that $gh = hg = e$. We usually write h as g^{-1}.

These rules at first are incredibly simple, but they allow for some interesting and beautiful structure to emerge. We will now look through a few examples of some common groups.

Example 2.1 (The integers with addition)**.** The set of integers $\mathbb{Z}$ with addition $+$ is a group.We write this as $(\mathbb{Z}, +)$. Addition on $\mathbb{Z}$ is associative since $(a + b) + c = a + (b + c)$ follows directly from the associativity of addition, which holds by construction of the integers. The identity is 0 as given any integer $a \in \mathbb{Z}$ $a + 0 = 0 + a = a$ $\forall a \in \mathbb{Z}$. For any $a \in \mathbb{Z}$, its inverse is simple $-a$ as $a + (-a) = (-a) + a = 0$

Other common examples of groups are $(\mathbb{R}, +), (\mathbb{C}, *)$ and $(\mathbb{Q}, *)$. These are all basic examples that can easily be proven.

Definition 2.2 (Abelian Groups)**.** A group G is an abelian group if for any $a, b \in G$:

$$ab = ba$$

All of the groups mentioned prior are examples of abelian groups. We will now look at non-abelian groups.

Example 2.2 (Symmetric groups)**.** Let X_n be the set of bijective functions from $\{1, 2, \ldots, n\}$ to itself. Combining this set with the operation of function composition $\circ$, we have the group $\mathcal{S}_n$. To explain this, we will focus on the group $\mathcal{S}_3$ which is the group of bijective functions from $\{1, 2, 3\}$ to itself. Consider an element $f \in \mathcal{S}_3$ where $f(1) = 2, f(2) = 3, f(3) = 1$. We can write this as: $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$ Or we can write it as (123). For the rest of this document, we will be writing elements of $\mathcal{S}_n$ in the second version presented (known as cycle notation). For the identity element of this group, it will simply be id_n, and for an arbitrary element $f \in \mathcal{S}_n$, the inverse is f^{-1}.

Example 2.3 (General Linear Group)**.** The set of square $n \times n$ invertible matrices under matrix multiplication is known as the General Linear group, denoted $GL(n, K)$, where K is a field and $n \in \mathbb{N}$.

Now that we have shown examples of abelian and non-abelian groups, we can move on to proving some basic facts about groups.

Lemma 2.1. Let G be a group. The following are true:

1. There exists at most 1 identity element
2. Every element $g \in G$ has a unique inverse g^{-1}
3. $(g^{-1})^{-1} = g$
4. $(ab)^{-1} = b^{-1}a^{-1}$
5. if $ab = ac$ then $b = c$
6. if $ba = ca$ then $b = c$

Proof of 1
Let G be a group with 2 identity elements e and $\hat{e}$. Since they are both identities, that means for any $g \in G$, $eg = ge = e$ and $g\hat{e} = \hat{e}g = g$. Now let $g = e$. Via the property of the identity: $e\hat{e} = \hat{e}e = e$ and $\hat{e}e = e\hat{e} = \hat{e}$. Via the property of transitivity, $e = \hat{e}$ $\square$

Proof of 2
Let $g \in G$ be given and let $a, b \in G$ be inverses for g. a and b are inverses for g which means $ga = ag = e$ and $gb = bg = e$. Consider agb. Using associativity, we can show that:

$$agb = (ag)b = eb = b$$

And

$$agb = a(gb) = ae = a$$

So $agb = a$ and $agb = b$, therefore $a = b$ implying the inverse element is unique. $\square$

Proof of 3
Let $g \in G$ and $g^{-1} \in G$ be given. By definition: $gg^{-1} = e$. Multiplying both right sides by $(g^{-1})^{-1}$ gets us:

$$gg^{-1}(g^{-1})^{-1} = e(g^{-1})^{-1} \implies$$

$$ge = (g^{-1})^{-1} \implies g = (g^{-1})^{-1}$$

Which confirms $(g^{-1})^{-1} = g$. $\square$

Proof of 4
let $a, b \in G$ and let $c \in G$ be the inverse of ab. This means:

$$(ab)c = e$$

Multiplying the left side by a^{-1}:

$$a^{-1}(ab) = a^{-1} \implies (a^{-1}a)bc = a^{-1} \implies bc = a^{-1}$$

If we now multiply the left hand side by b^{-1} we get:

$$b^{-1}bc = b^{-1}a^{-1} \implies (b^{-1}b)c = b^{-1}a^{-1} \implies c = b^{-1}a^{-1}$$

So we have $c = b^{-1}a^{-1}$ and since we defined c to be the inverse of (ab) i.e. $(ab)^{-1}$, we have shown that $(ab)^{-1} = b^{-1}a^{-1}$. $\square$

Proof 5 and 6 are exercises for the reader.

3 Subgroups

Definition 3.1 (Definition of a subgroup)**.** Let $(G, *)$ be a group. A subset $H \subseteq G$ is called a subgroup of G if H itself forms a group under the operation $*$ if H is a subgroup of G we write $H \leq G$.

For H to be a group under the same operation, it must inherit the group axioms. Since H lives inside G, it automatically takes associativity, and therefore only 3 rules must be verified to see if H is a subgroup of G.

In simple terms, $H \leq G$ if and only if:

1. Closure: $\forall a, b \in H,\ \ ab \in H$
2. Identity: The identity element $e \in G$ is also in H
3. Inverses $\forall g \in H,\ \ g^{-1} \in H$

Checking all of these axioms for every subgroup will become tedious. Instead, we use Theorems to help.

Theorem 3.1 (Subgroup test)**.** A non-empty subset $H \subseteq G$ is a subgroup ($H \leq G$) if and only if:

1. $\forall a, b \in H \implies ab \in H$
2. $\forall a \in H \implies a^{-1} \in H$

The proof in the $\implies$ direction is trivial, so we will focus on the proof in the $\impliedby$ direction. i.e., if a non-empty subset of G follows those 2 conditions, then it is a subgroup in G.

Let $a, b, c \in H$. Since $H \subseteq G$, these elements are also in G. Because the operation $*$ is associative for all elements in G, it is automatically associative for the elements in H. Closure and inverses are both guaranteed by the 2 conditions of the subgroup test. Since we know H is non-empty, it implies there exists one element $x \in H$. Condition 2 tells us that $x^{-1} \in H$ and condition 1 tells us that $xx^{-1} \in H$ and since $xx^{-1} = e$, then $e \in H$ and therefore H has an identity element. Since H has met the requirements of all of the group axioms, it must be a subgroup of G. i.e., $H \leq G$. $\square$

With this theorem, we can prove that groups are subgroups of others more quickly.

Example 3.1 (The integers are a subgroup of the real numbers)**.** $(\mathbb{Z}, +)$ is a subgroup of $(\mathbb{R}, +)$

For any group G, the trivial group $\{e\}$ and the group itself are always subgroups.

Definition 3.2 (The order of an element)**.** Let G be a group and let $g \in G$ be given. Suppose $g^n = e$ i.e. $\underbrace{gg \dots g}_{\text{n times}} = e$. The smallest positive integer n when this is true is called the order of g. We denote this $|g| = n$.

If there does not exist an n then we say that the order of g is infinite. Note that the only element to have order 1 is the identity element itself. Furthermore, the order of a *group* is simply the number of elements within it. A classic example of this can be the $90°$ rotation matrix A in the general linear group. the order of A would be 4. Another example is a reflection s which has order 2.

Definition 3.3 (Cyclic groups)**.** Let G be a group and let $g \in G$ be given. We define the *cyclic* group to be $\langle g \rangle = \{g^k \mid k \in \mathbb{Z}\}$. We call g the generator of this group.

We say a group is cyclic if there exists an element $g \in G$ such that $\langle g \rangle = G$. An example is the group of integers represented by addition, i.e. $\langle 1 \rangle = \mathbb{Z}$. Since 1 has infinite order, we say that this group is infinite. We can have multiple generators for 1 group.

Consider 2 elements a, b with orders n and m respectively; we can define a group generated by these 2 elements as:

$$\langle a, b\rangle = \{a^x b^y \mid x, y \in \mathbb{Z}^+\}$$

It is clear that any cyclic group is also abelian.

4 Homomorphisms

Homomorphisms are very useful in abstract algebra as they preserve the algebraic operation between 2 structures, allowing us to translate problems from one system to another where they might be easier to solve.

In group theory, we define a group in the following way:

Definition 4.1 (A group homomorphism)**.** Given 2 groups $(G, *)$ and $(H, \star)$, a homomorphism is a function $f : G \to H$ such that for any $a, b \in G$:

$$f(a * b) = f(a) \star f(b)$$

This definition essentially states that group operations are preserved between the 2 groups. With this definition, we can already prove some interesting consequences.

Lemma 4.1. Let G and H be groups. Let f be a homomorphism between G and H. The following are true:

1. $f(e_G) = e_H$
2. $\forall g \in G,\ f(g^{-1}) = f(g)^{-1}$
3. $\forall g \in G,\ \ f(g^n) = f(g)^n$
4. If $g \in G$ has finite order i.e. $|g| = n$ then the order of $|f(g)|$ must divide n

Proof of 1:

Let G and H be groups and let f be a homomorphism from G to H. Consider $f(e_G)$, since f is a homomorphism, it means:

$$f(e_G) = f(e_G * e_G) = f(e_G) \star f(e_G)$$

Which implies $f(e_G) = f(e_G) \star f(e_G)$. Multiplying both sides by $f(e_G)^{-1}$ will get us:

$$e_H = f(e_G)$$

□

Proof of 2:

Let $g \in G$. Because of (1) we have:

$$f(g) \star f(g^{-1}) = f(g * g^{-1}) = f(e_G) = e_H$$

This implies that $f(g) \star f(g^{-1}) = e_H$ which makes $f(g^{-1})$ the inverse of $f(g)$ which means:

$$f(g^{-1}) = f(g)^{-1}$$

□

Proof of 3:

Consider $f(g^n) = f(\underbrace{gg\dots g}_{\text{n times}}) = f(g) \star f(g) \star f(g) \star \cdots \star f(g) = f(g)^n$. □

Proof of 4:

Let $g \in G$ have order n and let $f(g)$ have order k. Because of (3), we know that $f(g^n) = f(g)^n$, which implies that

$$f(g)^n = f(g^n) = f(e_G) = e_H$$

Now using something called the *Division Algorithm* (this will be expanded upon further in later sections), we want to show that k must divide n, therefore we can write n in terms of k:

$$n = qk + r \quad \text{where } 0 \leq r < k$$

Substituting this back into our equation gets us:

$$e_H = f(g)^n = f(g)^{qk+r} = (f(g)^k)^q \star f(g)^r$$

Since we defined the order of $f(g)$ to be k, we can simplify to:

$$e_H = f(g)^r$$

However, this would imply that r is the order of $f(g)$ which would contradict our initial statement that being the order of $f(g)$ is equal to k. Therefore, $r = 0$ and thus:

$$n = qk \implies k \mid n$$

Proving our statement.

Definition 4.2 (Kernel of a homomorphism)**.** Let f be a group homomorphism between G and H. The *kernel* of f is the set:

$$\ker f = \{g \in G \mid f(g) = e_H\}$$

Lemma 4.2 (The kernel is a subgroup of G)**.** Let f be a group homomorphism between the groups G and H. First, we will show that if $a, b \in \ker f$, then

$ab \in \ker f$.

Let $a, b \in \ker f$ be given.

$$f(ab) = f(a) \star f(b) = e_H \star e_H = e_H$$

This shows that $f(ab) = e_H$ which therefore implies $ab \in \ker f$. Now consider $f(a^{-1})$. We can easily show:

$$f(a^{-1}) = f(a)^{-1} = e_H^{-1} = e_H$$

So $f(a^{-1}) = e_H$ which implies $a^{-1} \in \ker f$. Because $\ker f \subseteq G$ and because it satisfies the subgroup test, it must be a subgroup of G. $\square$

If we have a group G and a group homomorphism $f : G \to H$, we can now say that $\{e\}, G, \ker f$ are guaranteed to be subgroups of G.

Definition 4.3 (Image of a homomorphism)**.** Let f be a group homomorphism between G and H. The *image* of f is the set:

$$\text{Img } f = \{h \in H \mid h = f(g) \text{ for some } g \in G\}$$

Lemma 4.3 (The image is a subgroup of H)**.** The proof of this is similar to the proof in Lemma 1.3, so we will go through this quickly.

Let G and H be groups and let $f : G \to H$, be a homomorphism. Now consider $g, h \in \text{Img } f$. This means: $g = f(a)$ and $h = f(b)$ for some $a, b \in G$. If we have gh, we can rewrite this as:

$$gh = f(a)f(b) = f(ab)$$

Since $ab \in G$, this means there exists an element x in G such that $f(x) = gh$ which makes $gh \in \text{Img } f$. Now suppose we have $h \in \text{Img } f$. This implies $h = f(b)$ for some $b \in G$. Because $b \in G$, that means $b^{-1} \in G$. So consider $f(b^{-1}) = f(b)^{-1} = h^{-1}$ which implies $h^{-1} \in \text{Img } f$. Since $\text{Img } f \subseteq H$, it passes the subgroup test. It is a subgroup of H. $\square$

For the subgroup test, we must also show that the subgroup H is non-empty, for both the kernel and image of the homomorphism, we can use e_G and e_H respectively.

Lemma 4.4 (injective homomorphisms and kernels)**.** Let $f : G \to H$ be a homomorphism. Then f is injective if and only if $\ker f = \{e_G\}$.

First, we will prove that if f is injective then $\ker f = \{e_G\}$. Suppose: $f(x) = e_H$. We know that $f(x) = f(e_G)$ and since we defined f to be injective that means: $x = e_G$ i.e. $\ker f = \{e_G\}$.

Now for proving the converse, Let $\ker f = \{e_G\}$ and suppose $f(x) = f(y)$

for some $x, y \in G$. Using our previous lemmas and the fact that f is a homomorphism, we can show:

$$f(x) = f(y) \implies f(x)f(y)^{-1} = e_H \implies f(x)f(y^{-1}) = e_H \implies f(xy^{-1}) = e_H$$

This would mean $xy^{-1} \in \ker f$ therefore implying $xy^{-1} = e_G$. Now we can easily show that $x = y$ implying that f is injective. $\square$

5 Isomorphisms

Definition 5.1 (Isomorphisms)**.** Let G and H be groups. A function $f : G \to H$ is called an isomorphism if f is a bijective homomorphism. If such a function exists between G and H, then we say that G is isomorphic to H, and we can denote this $G \cong H$.

Isomorphisms are very important within abstract algebra as they preserve the algebraic properties from one group to another, which can become very useful in helping us solve problems, as we can simply move from more complicated structures to simpler ones.

Lemma 5.1. let G and H be groups. If $f : G \to H$ is an isomorphism then $f^{-1} : H \to G$ is also an isomorphism.

We know that if f is bijective, then f^{-1} is also bijective which means all we need to show is that f^{-1} is a homomorphism.

Let f^{-1} be a bijective function and let $a, b \in H$ be given. We know that f is an isomorphism, which means we can rewrite $a = f(x)$ and $b = f(y)$ for some $x, y \in G$. Consider $f^{-1}(ab)$:

$$f^{-1}(ab) = f^{-1}(f(x)f(y)) = f^{-1}(f(xy)) = xy$$

Now consider $f^{-1}(a)f^{-1}(b)$:

$$f^{-1}(a)f^{-1}(b) = f^{-1}(f(x))f^{-1}(f(y)) = xy$$

Which shows $f^{-1}(ab) = f^{-1}(a)f^{-1}(b)$ which means that f^{-1} is a homomorphism. Since f^{-1} is a bijective homomorphism, we have proved it is an isomorphism too. $\square$

Lemma 5.2. Let G, H, K be groups and let $p : G \to H$ and $f : H \to K$ be isomorphisms. Then $f \circ p$ is an isomorphism.

We know that if f and p are bijective functions, then $f \circ p$ is also bijective, which means we need to prove that $f \circ p$ is a homomorphism. Let $a, b \in G$. Consider $f \circ p(ab)$:

$$f \circ p(ab) = f(p(ab)) = f(p(a)p(b)) = f(p(a))f(p(b)) = f \circ p(a) f \circ p(b)$$

Therefore showing that $f \circ p$ is a homomorphism. Since it is a bijective homomorphism, we can conclude that $f \circ p$ is an isomorphism. $\square$

Theorem 5.1 (Isomorphisms are equivalence relations)**.** This theorem is incredibly trivial to prove due to the previous lemmas.

Let G be a group. Using the identity map, we can say that G is isomorphic to itself i.e. $G \sim G$.

If $G \cong H$, that implies there exists an isomorphism $f : G \to H$. Due to lemma 1.6, we know that f^{-1} is also an isomorphism which means: $H \cong G$ i.e. $G \sim H \implies H \sim G$

Finally if $G \cong H$ and $H \cong K$ where G, H, K are groups, then there exists isomorphisms $f : G \to H$ and $p : H \to K$. We can use the isomorphism: $p \circ f : G \to K$ to show that $G \cong K$ i.e. $G \sim H,\ H \sim K \implies G \sim K$. Proving that isomorphisms are equivalence relations. $\square$

Theorem 1.2 can be applied to general isomorphisms that aren't groups, such as rings, fields, and vector spaces. As a result, Theorem 1.2 will not be repeated in later sections.

Lemma 5.3. Let $G \cong H$. If G is abelian, then H is abelian.

Let $a, b \in G$ be given. Consider $f(ab)$:

$$f(ab) = f(a)f(b)$$

And

$$f(ab) = f(ba) = f(b)f(a)$$

Which implies $f(a)f(b) = f(b)f(a)$. Let $x, y \in H$. Since f is an isomorphism, that means it is surjective which means $x = f(a)$ and $y = f(b)$. This means:

$$xy = f(a)f(b) = f(b)f(a) = yx$$

Which implies $xy = yx$ for any values of H, making it an abelian group. $\square$

Lemma 5.4. $G \cong H$ via some isomorphism f. let $a \in G$ and $|a| = n$, then $|f(a)| = n$.

Recall that $f(a)^n = f(a^n) = e_H$. Suppose $f(a)^m = e_H$ for some $m < n$. Using previous lemmas, we can show that: $f(a)^m = f(a^m)$, which implies $f(a^m) = e_H$. Since f is injective, this means $\ker f = \{e_G\}$ which would lead to: $a^m = e_G$. However, this contradicts our claim that $|a| = n$ so m cannot exist. $\square$

Example 5.1. Consider the groups $G = (\mathbb{R}, +)$ and $H = (\mathbb{R}^+/\{0\}, \times)$. An isomorphism between the two would be $f : G \to H$ defined by $f(x) = e^x$. We can easily see that:

$$f(x + y) = e^{x+y} = e^x e^y = f(x)f(y)$$

Is an isomorphism. Note that e^x is also a bijective function from $\mathbb{R}$ to $\mathbb{R}^+/\{0\}$. The exponential map can also be used to send Lie algebras to Lie groups, which we will look at later in this series.

6 Cosets

Definition 6.1 (left cosets)**.** Let G be a group and $H \leq G$. A *left coset* is the set: $gH = \{gh : h \in H\}$

Example 6.1. Let $G = \mathbb{Z}_8$. This is the set of elements $\{0, 1, 2, 3, 4, 5, 6, 7\}$ with the group operation $\oplus$ (addition modulo 8 in this instance). A valid subgroup of $\mathbb{Z}_8$ would be the group $S = \{0, 2, 4, 6\}$. This is actually a cyclic group generated by 2, so $S = \langle 2 \rangle$. Checking this is a coset is left as an exercise. A valid left coset of H is $1 \oplus \langle 2 \rangle = \{1, 3, 5, 7\}$.

A right coset Hg is defined just like a left coset, just with right-hand multiplication.

Cosets are incredibly useful, as we will soon find out they partition the group G and are used in the definition of *normal* subgroups as well as being used in a very important theorem.

Lemma 6.1. *Let G be a group and let $H \leq G$. if $x \in yH$ then $xH = yH$.*

Let $x \in yH$ be given. This means that $x = yh$ for some $h \in H$. We will call this arbitrary value h_1. Now, suppose we have $a \in xH$. We can write $a = xh_2$. Substituting our value for x, we get $a = (yh_1)h_2 = y(h_1h_2)$ which would imply that $a \in yH$. Thus, we have proved that $xH \subseteq yH$. let $b \in yH$ be given. This would mean $b = yh_3$. Using our equation for x, we can rewrite y as $y = xh_1^{-1}$. Substituting this in gets us: $b = (xh_1^{-1})h_3 = x(h_1^{-1}h_3)$ which implies $b \in xH$. Now we have proved that $yH \subseteq xH$. Since we have shown that $xH \subseteq yH$ and $yH \subseteq xH$, it must mean that $xH = yH$. $\square$

Lemma 6.2. *Let G be a group and $H \leq G$. Suppose we have xH and yH as left cosets. Then they are either equal or disjoint.*

To prove this, we will assume that $xH \neq yH$ and that there exists an element $z \in xH \cap yH$. Since $z \in xH \cap yH$, this implies $z \in xH$ and $z \in yH$. Via Lemma 1.10, this would imply that $zH = xH$ and $zH = yH$, which further implies $xH = yH$, contradicting our previous assumption that $xH \neq yH$. Therefore $xH \cap yH = \varnothing$ or $xH = yH$. $\square$

Lemma 6.3. *Let G be a group and $H \leq G$. Then every element $g \in G$ belongs to some left coset.*

Let $g \in G$. We know that $e \in H$ as it is a subgroup. This means we can write $g = ge$, which would further imply that $g \in gH$ for any arbitrary value of g.

Theorem 6.1 (Left cosets of H partition G)**.** *Let G be a group and $H \leq G$. Then the left cosets partition G.*
Lemma 1.12 implies that every element of G lives in some left coset, and Lemma

1.11 says that 2 cosets are equal or disjoint, therefore implying that cosets partition G. $\square$

Due to Theorem 1.3, we can write G as:

$$G = \bigcup_{i=1}^{n} gH_i$$

In this instance, we use $n = [G : H]$ to denote the number of left cosets of H. Because the left cosets partition G, we can also say that:

$$|G| = \sum_{i=1}^{n} |gH_i| = |gH_1| + |gH_2| + \cdots + |gH_n|$$

Note that G must be finite for this to work

Lemma 6.4. Let G be a group and $H \leq G$. Then $|gH| = |H|$ for any $g \in G$.

Consider the function $f : H \to gH$ defined as $f(x) = gx$. We will show that this is a bijection.

Assume $f(x) = f(y)$, which implies $gx = gy$. Multiplying both sides on the left by g^{-1} gets us $x = y$, which implies f is injective. Now consider some arbitrary element $gh \in gH$. Then $h \in H$ and $f(h) = gh$. So gh has a preimage under f, hence f is surjective. Since f is injective and surjective, it is bijective, which implies $|H| = |gH|$. $\square$

7 Lagrange's Theorem

We now have all of the pieces to prove Lagrange's theorem.

Theorem 7.1 (Lagrange's Theorem)**.** Let G be a finite group and $H \leq G$. Then:

$$|G| = [G : H] \times |H|$$

Remember that we can write $|G|$ as

$$|G| = \sum_{i=1}^{n} |gH_i| = |gH_1| + |gH_2| + \cdots + |gH_n|$$

Since we know $|gH| = |H|$, we can rewrite this as:

$$|G| = \sum_{i=1}^{n} |H| = n|H|$$

Letting $n = [G : H]$ gets us:

$$|G| = [G : H] \times |H|$$

As required. $\square$

This theorem allows us to compute the number of left cosets of a subgroup H. This theorem also gives another interesting result.

Theorem 7.2 (Groups of prime order are cyclic)**.** Let G be a finite group of prime order p. Then G is cyclic.

Since $|G| = p$, this means $|H|$ is either 1 or p. take $h \in G$ and consider $\langle h \rangle \leq G$. Due to Lagrange's, this means $|\langle h \rangle|$ divides p. If $h = e$ then $|\langle h \rangle| = 1$. Let's assume $h \neq e$. So $|\langle h \rangle| \neq 1$ This would mean that $|\langle h \rangle| = p = |G|$ which means $\langle h \rangle = G$ therefore proving G is cyclic. $\square$

Lemma 7.1. Let G be a group. If G is cyclic, then it is an abelian group.

G is cyclic, implying that $G = \{g^k : k \in \mathbb{Z}\}$ for some $g \in G$. Let $a, b \in G$. We can rewrite them as $a = g^n$ and $b = g^m$ for some $n, m \in \mathbb{Z}$. Now we can perform:

$$ab = g^n g^m = g^{n+m} = g^{m+n} = g^m g^n = ba$$

Which shows $ab = ba$ for any values of $a, b \in G$, therefore proving G is abelian. $\square$

Theorem 7.3 (Groups of prime order are abelian)**.** Let G be a finite group of prime order. Then G is abelian.

This is trivial to prove. Using Theorem 1.5, we can say that any group of prime order is cyclic, and Lemma 1.14 says that any cyclic group is abelian. Combining the 2 tells us that any group of prime order is abelian. $\square$

8 Normal Subgroups

Definition 8.1 (Normal subgroup)**.** Let G be a group and let $H \leq G$. We say H is a normal subgroup of G if $gH = Hg \ \forall g \in G$. i.e.all left and right cosets are equal to each other. We can also write this as $gHg^{-1} = H$ which tends to be how it is formally introduced. If H is a normal subgroup, we denote it $H \trianglelefteq G$

Proving a subgroup is normal involves first proving $gHg^{-1} \subseteq H$ and $H \subseteq gHg^{-1}$. However,if we assume $gHg^{-1} \subseteq H$ and substitute g^{-1} in the position of g, we get: $g^{-1}Hg \subseteq H$. Moving the g's to the other side gets us: $H \subseteq gHg^{-1}$. This implies all we need to do to prove that a subgroup is normal is prove that $gHg^{-1} \subseteq H$.

Lemma 8.1. Given a group G, the following subgroups are normal:

1. The trivial group $\{e\}$
2. The whole group G

Proof of (1). Let $geg^{-1} \in g\{e\}g^{-1}$ be given. Using deduction, we can show that:

$$geg^{-1} = (ge)g^{-1} = (g)g^{-1} = gg^{-1} = e$$

We know $e \in H$ so therefore $g\{e\}g^{-1} \subseteq \{e\}$ which means $\{e\}$ is normal. $\square$

Proof of 2:

Let $aga^{-1} \in aGa^{-1}$ be given. We know that a and a^{-1} are in G so this means $aga^{-1} \in G$. Therefore G is normal. $\square$

Lemma 8.2 (The kernel is a normal subgroup)**.** Let G be a group and f be a homomorphism. Then $\ker f \trianglelefteq G$

Let $hgh^{-1} \in h \ker(f)\, h^{-1}$. Applying f to this element gets us:

$$f(hgh^{-1}) = f(h)f(g)f(h^{-1}) = f(h)e_H f(h^{-1}) = f(h)f(h^{-1}) = f(hh^{-1}) = f(e_G) = e_H$$

Which implies $hgh^{-1} \in \ker f$, which means $\ker f \trianglelefteq G$. $\square$

Lemma 8.3. Let G be an abelian group. Then every subgroup of H is normal.

Let H be a subgroup and consider $ghg^{-1} \in gHg^{-1}$. Because G is abelian, we can:

$$ghg^{-1} = gg^{-1}h = eh = h$$

And we know $h \in H$, therefore $gHg^{-1} \subseteq H$ implying H is normal.

9 Quotient Groups

Definition 9.1 (Quotient group)**.** Let G be a group and let $H \trianglelefteq G$. The *quotient group* denoted G/H is the set of left cosets with coset multiplication i.e. let $aH, bH \in G/H$. We define this multiplication as: $(aH)(bH) = (ab)H$.

Upon seeing this definition, people may wonder why H has to be normal. The reason is that coset multiplication must be *well-defined*: since a coset aH can have multiple representatives, we need the multiplication to be independent of which representative we choose. Suppose $aH = a'H$ and $bH = b'H$, so $a' = ah_1$ and $b' = bh_2$ for some $h_1, h_2 \in H$. Then:

$$a'b' = ah_1bh_2 = ab(b^{-1}h_1b)h_2$$

Since $H \trianglelefteq G$, normality gives us $b^{-1}h_1b \in H$, and therefore $(b^{-1}h_1b)h_2 \in H$. This means $a'b' \in abH$, i.e. $(a'b')H = (ab)H$, so the multiplication is well-defined.

Lemma 9.1 (G/H is a group)**.** Let G be a group and $H \trianglelefteq G$. Then G/H is a group under coset multiplication.

Associativity follows directly from associativity in G:

$$((aH)(bH))(cH) = (ab)H(cH) = ((ab)c)H = (a(bc))H = (aH)(bcH) = (aH)((bH)(cH))$$

The identity element is $eH = H$ since $(aH)(eH) = (ae)H = aH$. Finally, the inverse of aH is $a^{-1}H$ since $(aH)(a^{-1}H) = (aa^{-1})H = eH$. Hence G/H is a group. $\square$

10 The First Isomorphism Theorem

Theorem 10.1 (First Isomorphism Theorem)**.** Let $f : G \to H$ be a group homomorphism, then the map:

$$\varphi : G/\ker f \to \text{Img } f \quad \text{defined by } \varphi(gK) = f(g) \text{ where } K = \ker f$$

Is an isomorphism

Well-defined: Suppose $gK = g'K$, then $g^{-1}g' \in K = \ker f$, so $f(g^{-1}g') = e_H$, which implies $f(g)^{-1}f(g') = e_H$, therefore $f(g) = f(g')$, i.e. $\varphi(gK) = \varphi(g'K)$.

Homomorphism: Let $aK, bK \in G/K$. Then:

$$\varphi(aK \cdot bK) = \varphi((ab)K) = f(ab) = f(a)f(b) = \varphi(aK)\varphi(bK)$$

Injective: Suppose $\varphi(gK) = \varphi(g'K)$, then $f(g) = f(g')$, which implies $f(g)^{-1}f(g') = e_H$, so $f(g^{-1}g') = e_H$, meaning $g^{-1}g' \in K$, therefore $gK = g'K$.

Surjective: Let $h \in \text{Img } f$. By definition, $h = f(g)$ for some $g \in G$, so $\varphi(gK) = f(g) = h$.

Since φ is a bijective homomorphism, it is an isomorphism, and therefore $G/\ker f \cong \text{Img } f$. $\square$

11 Group Actions

Many groups seem extremely abstract and empty when viewed as just groups; however, their true nature is revealed when we apply these groups to arbitrary sets of mathematical objects.

Definition 11.1 (Group action)**.** Let G be a group and let X be a set. A group action is a function:

$$\cdot : G \times X \to X$$

Which satisfies the following 2 axioms:

1. $e \cdot x = x,\ e \in G, \forall x \in X$
2. $g \cdot (h \cdot x) = (gh) \cdot x$, $\forall x \in X$, $\forall g, h \in G$

Example 11.1 (Triangle)**.** Let X be the vertices of the superior shape (equilateral triangle) $X = \{v_1, v_2, v_3\}$. We can define a group action using $G = D_3$, which represents the symmetries of a triangle. The group action would therefore be:

$$D_3 \times X \to X$$

This will be the example we use for the rest of the section. We can show it's a group action by showing it satisfies the axioms:

$e \cdot v_i = v_i, \ i \in \{1, 2, 3\}$.

The other axiom is an exercise for the reader.

Specifically, this example is interesting as the group action gives a homomorphism to $\mathcal{S}_3$

Definition 11.2 (Orbits)**.** Given a group G and a set X, the orbit of an element $x \in X$ is defined as:

$$\mathrm{Orb}(x) = \{g \cdot x : g \in G\}$$

A group action is transitive if and only if $\mathrm{Orb}(x) = X$ for some $x \in X$

Definition 11.3 (Stabilisers)**.** Given a group G and a set X, the stabiliser of an element $x \in X$ is defined as:

$$\mathrm{Stab}(x) = \{g \in G : g \cdot x = x\}$$

We say the action of G on X is free if and only if $\mathrm{Stab}(x) = \{e\}$, $\forall x \in X$

Lemma 11.1. Let G be a group and X be a set. Then $\mathrm{Stab}(x)$ is a subgroup of G.

We know that $\mathrm{Stab}(x) \neq \varnothing$ as $e \in \mathrm{Stab}(x)$. Now consider $g, h \in \mathrm{Stab}(x)$. Applying $(gh) \cdot x$ gets us:

$$(gh) \cdot x = g \cdot (h \cdot x) = g \cdot x = x$$

Which implies $gh \in \mathrm{Stab}(x)$. Now let $g \in \mathrm{Stab}(x)$. We know that: $g \cdot x = x$. Apply g^{-1} to both sides of the equation:

$$g^{-1} \cdot (g \cdot x) = g^{-1} \cdot x \implies (g^{-1}g) \cdot x = g^{-1} \cdot x \implies e \cdot x = g^{-1} \cdot x = x$$

Which implies $g^{-1} \in \mathrm{Stab}(x)$. Therefore, implying that $\mathrm{Stab}(x) \leq G$. □

Lemma 11.2. Let G be a group and let $H = \mathrm{Stab}(x)$. Then G/H is bijective to $\mathrm{Orb}(x)$ via the map: $\varphi(gH) = g \cdot x$.

First we will show that this map is well defined i.e. $gH = kH \implies \varphi(gH) =$

$\varphi(kH)$. Suppose $gH = kH$. This means that $g = kh_1$ for some $h_1 \in H$. Now consider $\varphi(gH) = g \cdot x$. Using $g = kh_1$, we can show that:

$$\varphi(gH) = g \cdot x = (kh_1) \cdot x = k \cdot (h_1 \cdot x) = k \cdot x = \varphi(kH)$$

Which implies φ is well defined.

Showing that φ injective is a trivial calculation. Assume $\varphi(gH) = \varphi(kH)$, then:

$$g \cdot x = k \cdot x \implies k^{-1} \cdot (g \cdot x) = k^{-1} \cdot (k \cdot x) \implies (k^{-1}g) \cdot x = (k^{-1}k) \cdot x = x$$

This implies that $k^{-1}g \in H$ which implies $gH = kH$, proving it is injective. Let $y \in \mathrm{Orb}(x)$. This means there exists $g \in G$ such that $y = g \cdot x$. We can use gH as our input, proving φ is surjective.

Since we have proved our function is injective and surjective, we have proved it is bijective, which means we can say $|\mathrm{Orb}(x)| = [G : \mathrm{Stab}(x)]$. $\square$

Theorem 11.1 (Orbit-Stabiliser Theorem)**.** Let G be a group and let X be a set. Then:

$$|G| = |\mathrm{Orb}(x)| \times |\mathrm{Stab}(x)|$$

This result can be proved via Lagrange's Theorem, Lemma 11.1 and 11.2.

Lemma 11.1 shows that $\mathrm{Stab}(x) \leq G$ which means:

$$|G| = [G : \mathrm{Stab}(x)] \times |\mathrm{Stab}(x)|$$

Via Lagrange's theorem. Lemma 11.2 shows that $|\mathrm{Orb}(x)| = [G : \mathrm{Stab}(x)]$ and combining the 2 together gives us:

$$|G| = |\mathrm{Orb}(x)| \times |\mathrm{Stab}(x)|$$

Just like we wanted. $\square$